

W4: COSY INFINITY Cross-Validation of FELsim Beamline Optimisation

Eremey Valetov

15 February 2026

1 Objective

Cross-validate FELsim’s 11-stage sequential Nelder-Mead beamline optimisation against COSY INFINITY’s internal FIT command. Both codes solve the same problem: matching Twiss parameters at the undulator entrance of the UH MkV FEL transport line (118 elements, 23 quadrupoles) to the targets of Weinberg, Fisher & Li [1].

The goal is independent confirmation that the optimised quadrupole settings achieve the required undulator matching — not necessarily identical currents, but equivalent final Twiss.

2 Method

2.1 Translation to COSY FIT

FELsim’s 11-stage optimisation was translated into 11 sequential COSY FIT blocks within a single FOX input file. Each FIT block:

- Declares variables (quadrupole currents) and bounds,
- Re-executes the full lattice (via `LATTICE`),
- Evaluates a combined MSE objective from the Twiss/dispersion goals,
- Iterates until convergence ($\varepsilon = 10^{-8}$, $N_{\max} = 1000$).

Optimised variables persist between FIT blocks, so each subsequent stage inherits the results of all prior stages.

2.2 Key Implementation Details

Combined MSE objective. COSY’s FIT with individual objectives (`ENDFIT` with multiple `OBJ`) resets to initial values when *any* objective fails to converge. A single combined MSE objective avoids this:

$$\text{OBJ}(1) = \frac{1}{N} \sum_{i=1}^N w_i (f_i - g_i)^2, \quad (1)$$

where f_i is the computed Twiss parameter, g_i the target, and w_i the weight. The sum is evaluated without `CONS()`, preserving DA derivatives for gradient-based optimisation (FIT algorithm 1).

Initial Twiss from beam distribution. FELsim computes Twiss from particle statistics of a Gaussian beam ($\sigma_x = 0.8$ mm, $\varepsilon = 0.101 \pi$ mm mrad). The initial beta function is $\beta_0 = \sigma_x^2/\varepsilon \approx 6.34$ m, not $\beta_0 = 1$ m. Using the wrong initial Twiss gives completely incorrect transport ($\alpha_x = -2.8$ instead of ≈ 0 after Stage 1). This was the most critical implementation finding.

Dipole model. Hard-edge dipoles (DIL without FC) with edge angles from the beamline specification. The DIL β parameter is set to zero (no fringe field integral), meaning edge kicks use $\tan(\epsilon)/\rho$ without the fringe field correction term. See Section 4 for implications.

3 Results

3.1 Beam and Target Parameters

Table 1: Beam parameters and undulator Twiss targets.

Parameter	Value
Energy	40 MeV
Normalised emittance ε_n	8π mm mrad
Geometric emittance ε	0.101π mm mrad
Initial beam size $\sigma_{x,y}$	0.8 mm
Initial β_0	6.34 m
Target β_x	1.400 m
Target α_x	0.470
Target β_y	0.242 m
Target α_y	0.000

These parameters are identical for the S1 (2 ps) and S3 (0.5 ps) configurations — the transverse beam parameters and undulator Twiss targets do not depend on bunch length, so the optimisation result is the same for both.

3.2 COSY FIT Convergence

The full 11-stage FIT converges in a single COSY execution. Table 2 compares the final Twiss at the undulator entrance.

Table 2: Final Twiss at undulator entrance: COSY FR 0, COSY FR 1, FELsim, and targets.

Parameter	Target	COSY FR 0	COSY FR 1	FELsim NM
β_x (m)	1.400	1.400	1.400	1.400
α_x	0.470	0.470	0.470	0.470
β_y (m)	0.242	0.242	0.242	0.242
α_y	0.000	0.000	0.000	0.000
MSE	—	4.50e-9	7.93e-8	2.89e-5

Both COSY runs match the targets to high precision. COSY’s gradient-based FIT (algorithm 1, using DA derivatives) converges $\sim 10^4 \times$ more precisely than Nelder-Mead. FR 0 and FR 1 achieve

comparable MSE ($\sim 10^{-8}$ – 10^{-9}), confirming that fringe fields do not degrade convergence when warm-starting is used.

3.3 Quadrupole Current Comparison

A detailed comparison of optimised quadrupole currents across FELsim, COSY FR0, and COSY FR1 is given in Table 4 (Section 3.6).

Stages 1–2 and 6–9 show excellent agreement across all three runs ($\Delta < 0.1$ A). The principal discrepancy is Stage 5 (double triplet, elements 33–43): COSY finds a local minimum with *negative* currents for elements 35 and 37, while FELsim (constrained to $[0, 10]$ A) finds all-positive currents. A four-point multistart over Stage 5 starting values confirmed that COSY’s gradient-based optimiser consistently converges to the negative-polarity basin.

Stages 10 and 11 also differ from FELsim, partly because the different Stage 5 outcome propagates through the downstream lattice, and partly due to the dipole model difference discussed in Section 4.

3.4 Dispersion

The COSY-optimised solution shows residual horizontal dispersion $\eta_x \approx 33$ mm at the undulator entrance. The Stage 11 objective includes dispersion zeroing at element 92 (5th chromaticity quad) with weight 0.5, but does not constrain η'_x . Between element 92 and the undulator (element 117), the non-zero η'_x causes dispersion to accumulate through the final triplet drifts.

FELsim shows comparable residual dispersion at element 92 (0.017 in optimizer units), confirming this is a lattice design feature rather than a code discrepancy.

3.5 Single-Dipole Transfer Matrix Comparison

To quantify the dipole model difference, we compare the 6×6 transfer matrix of individual DPW-DPW “sandwiches” between FELsim and COSY under three fringe settings: FR 0 (no fringe), FR 3 (third-order fringe), and FR 3 with FC (Enge coefficients, where available).

Table 3: Vertical edge kick M_{43} and Y-plane stability for representative dipole triplets. FELsim uses a triangle-model fringe correction; COSY uses the DIL element under different FR settings.

Dipole	θ (°)	FELsim	COSY FR 0	COSY FR 3	COSY FR 3+FC
M_{43} (<i>vertical kick</i> , rad/m)					
Transport	1.5	-7.42×10^{-3}	-7.71×10^{-3}	-6.78×10^{-3}	—
Chicane	4.0	-1.11×10^{-1}	-1.21×10^{-1}	-8.93×10^{-2}	—
FC1	11.25	-9.50×10^{-1}	-1.04	-7.79×10^{-1}	-2.27×10^{-1}
$ \text{Tr}(M_y)/2 $ (<i>stability</i> , < 1 required)					
Transport	1.5	0.9997	0.9997	0.9997	—
Chicane	4.0	0.9977	0.9975	0.9982	—
FC1	11.25	0.9823	0.9805	0.9855	0.9957

Three findings emerge from Table 3:

1. FELsim’s Y-kick lies between COSY FR 0 and FR 3 for every dipole type, but closer to FR 0 (the hard-edge model).

2. The discrepancy grows with bending angle: 3.9% at 1.5° , 8.9% at 4° , and 10% at 11.25° (FR0 vs FELsim).
3. The FC Enge coefficients (available for the FC1/FC2 chicane dipoles) produce much weaker Y-kicks (76% less than FELsim), indicating that realistic fringe fields partially cancel the geometric edge focusing.

The cumulative effect of all 14 dipole sandwiches in FELsim gives $|\text{Tr}(M_y)/2| = 1.203$ — formally *unstable* without quadrupoles. The quads must therefore provide sufficient Y-plane focusing to stabilise the beamline. When FELsim-optimised currents (calibrated to FELsim’s dipole model) are injected into COSY (with its different edge kicks), the compensation breaks down, explaining the Y-plane instability observed in cross-validation ($|\text{Tr}(M_y)/2| = 29$).

3.6 COSY FIT with First-Order Fringe Fields (FR 1)

To assess the impact of dipole fringe fields on the optimisation, we re-ran the full 11-stage COSY FIT with the FR 1 command enabled, which includes first-order fringe field effects (Enge-model based Runge-Kutta integration through the fringe region).

Cold-start failure. Running FR 1 from the same default initial guesses as FR 0 produces poor convergence: $\text{MSE} = 1.7 \times 10^{-1}$ at order 3 and 5.7×10^{-1} at order 1. The Y-plane becomes unstable ($|\text{Tr}(M_y)/2| = 1.43$) because the changed dipole edge kicks push the sequential FIT stages into incompatible local minima. The gradient-based FIT (algorithm 1) cannot recover once early stages converge to a wrong basin.

Warm-start from FR 0. Using the converged FR 0 quadrupole currents as initial guesses for FR 1 resolves the convergence problem completely. The optimiser needs only small adjustments from the nearby FR 0 solution:

Run	MSE	ΔI_{max} (A)
FR 0 (baseline)	4.50×10^{-9}	—
FR 1 (cold start)	1.66×10^{-1}	—
FR 1 (warm from FR 0)	7.93×10^{-8}	0.76

The warm-started FR 1 run achieves $\text{MSE} = 7.9 \times 10^{-8}$, comparable to the FR 0 baseline, with a maximum current change of 0.76 A. Stages 1–2 and 6–9 barely change; the main adjustments are in the triplets (Stages 3, 5, 10–11).

Table 4 compares the quadrupole currents across all three converged runs: FELsim, COSY FR 0, and COSY FR 1.

FR 3 computational cost. Third-order fringe field computation (FR 3) was attempted but is prohibitively expensive: each lattice evaluation takes >15 minutes due to Runge-Kutta integration through 14 dipole fringe regions, even at first-order DA. A full 11-stage FIT (requiring hundreds of lattice evaluations) would take days. The single-dipole comparison (Table 3) provides FR 3 results for individual elements.

Table 4: Optimised quadrupole currents: FELsim, COSY FR 0, COSY FR 1 (warm-started from FR 0), and RF-Track (Stage 11 only, warm-started from FELsim).

Elem	Label	Section	F/D	FELsim (A)	COSY FR 0 (A)	COSY FR 1 (A)	RFT-opt (A)	Stage
1	LQ1	First doublet	F	0.822	0.839	0.842	—	1
3	LQ2	First doublet	D	1.043	1.052	1.057	—	1
10	DPHQ	Chromaticity 1	F	3.883	3.957	3.979	—	2
16	F1QA	Triplet 1	D	2.240	1.993	1.994	—	3
18	F1QB	Triplet 1	F	4.953	4.859	4.955	—	3
20	F1QC	Triplet 1	D	3.426	3.499	3.599	—	3
27	DPHQ	Chromaticity 2	F	4.666	4.685	4.711	—	4
33	F2QA	Double triplet	D	2.694	0.357	0.518	—	5
35	F2QB	Double triplet	F	2.652	−1.975	−1.849	—	5
37	F2QC	Double triplet	D	0.277	−1.995	−1.985	—	5
50	DPHQ	Chromaticity 3	F	4.674	4.685	4.696	—	6
56	F3QA	IP doublet	D	3.122	3.168	3.218	—	7
58	F3QB	IP doublet	F	3.313	3.321	3.361	—	7
61	F3QC	Post-IP doublet	F	5.178	5.166	5.252	—	8
63	F3QD	Post-IP doublet	D	4.043	4.011	4.080	—	8
70	DPHQ	Chromaticity 4	F	4.682	4.662	4.811	—	9
76	F4QA	Triplet 2	D	3.934	3.869	3.953	—	10
78	F4QB	Triplet 2	F	4.079	4.452	4.506	—	10
80	F4QC	Triplet 2	D	0.014	0.608	0.592	—	10
87	DPHQ	Chromaticity 5	F	1.362	1.259	1.610	2.045	11
93	UBQ1	Final triplet	D	0.945	2.348	2.473	1.405	11
95	UBQ2	Final triplet	F	2.885	3.802	3.602	0.722	11
97	UBQ3	Final triplet	D	2.192	4.747	3.995	0.111	11

4 Physics Model Differences

4.1 Dipole Edge Fringe Fields

FELsim’s `dipole_wedge` class applies a simplified triangle-model fringe field correction to the vertical edge kick:

$$T_y = \tan\left(\epsilon - \frac{\ell_e}{6} \frac{1 + \sin^2\epsilon}{\rho \cos\epsilon}\right), \quad (2)$$

where ℓ_e is the physical DPW length, ϵ is the edge angle, and ρ is the bending radius. COSY’s DIL with $\beta = 0$ (our FR 0 baseline) gives $T_y = \tan\epsilon$ without fringe correction, while FR 3 computes fringe fields to third order using the half-gap parameter.

Each individual dipole shows a 4–20% difference in the Y-kick (Table 3). Over 14 dipoles (28 edge kicks), this accumulates to Y-plane instability when currents are exchanged between codes.

COSY provides more physically accurate fringe field treatments: the `FC` command with Enge function coefficients (measured values are available for the FC1/FC2 dipoles), the `FR` command for higher-order fringe fields, and the `MGE` command for 1D field map integration. The `MGE` approach requires a correctly scaled field map (currently unavailable for this beamline).

4.2 Current Sign Handling

FELsim’s quadrupole model uses $k = |eGI/(m_e c \beta \gamma)|$, making the transfer matrix independent of the sign of I . COSY’s MQ uses $B_{\text{pole}} = \pm G I r$, where the sign determines the focusing plane. Con-

sequently, COSY’s FIT can explore negative-current solutions that reverse the focusing/defocusing planes, while FELsim’s Nelder-Mead (with bounds $[0, 10]$ A) is restricted to the positive quadrant.

This explains why COSY consistently finds the negative-polarity Stage 5 solution: it is a genuine local minimum accessible to COSY but not to FELsim’s bounded optimiser. Both solutions satisfy the Stage 5 objectives.

5 Summary

1. COSY INFINITY FIT successfully reproduces the 11-stage beamline optimisation in a single execution, achieving $\text{MSE} \sim 10^{-8}$ – 10^{-9} for both FR 0 and FR 1 (warm-started). FELsim’s Nelder-Mead achieves $\text{MSE} = 2.9 \times 10^{-5}$.
2. All three codes/settings match the undulator Twiss targets to high precision. The lower COSY MSE reflects the advantage of gradient-based (DA) optimisation over Nelder-Mead.
3. Enabling first-order fringe fields (**FR 1**) requires warm-starting from the FR 0 solution. Cold-start with FR 1 fails ($\text{MSE} \sim 0.2$) because the changed dipole edge kicks push the sequential FIT stages into incompatible local minima.
4. Third-order fringe fields (**FR 3**) are prohibitively expensive: each lattice evaluation takes >15 minutes due to Runge-Kutta integration through 14 dipole fringe regions. Single-dipole comparisons (Table 3) quantify the FR 3 effect.
5. The principal discrepancy is Stage 5, where COSY converges to negative-polarity currents while FELsim (constrained to $[0, 10]$ A) finds all-positive currents (Section 4).
6. Initial Twiss from the beam distribution ($\beta_0 = \sigma^2/\varepsilon \approx 6.34$ m) is critical for correct transport computation.
7. Direct current injection between codes fails due to the dipole edge fringe correction difference (5–8% per edge, accumulating to Y-plane instability). The codes agree on final Twiss when each optimises within its own model.
8. S1 and S3 produce identical transverse optimisation results.

Key enablers for successful optimisation. Several design choices were critical for making the 11-stage optimisation converge reliably:

1. **Sequential decomposition into physically motivated stages.** The 23-quadrupole problem is decomposed into 11 stages, each corresponding to a distinct optical section (doublet, triplet, chromaticity section). Each stage optimises 1–4 variables against local objectives, avoiding the curse of dimensionality that would afflict a monolithic 23-variable optimisation.
2. **Joint 4-variable final stage.** With the asymmetric undulator Twiss targets from the paper ($\beta_x \neq \beta_y$, $\alpha_x \neq 0$), the final triplet alone (3 variables) is overconstrained against 4 Twiss objectives. Promoting chromaticity quad 5 (element 87) to a 4th joint variable resolves this: MSE drops from 8.6×10^{-2} (3-var) to 4.3×10^{-9} (4-var). A differential-evolution global search confirms this is the global optimum.

3. **Correct initial Twiss from the beam distribution.** The initial beta function must be computed from the beam statistics, $\beta_0 = \sigma_x^2/\varepsilon \approx 6.34$ m, not the common default $\beta_0 = 1$ m. Using the wrong initial Twiss propagates through the entire lattice and produces completely incorrect transport (e.g., $\alpha_x = -2.8$ instead of ≈ 0 after Stage 1).
4. **Combined MSE objective for gradient-based optimisation.** COSY’s FIT with individual objectives resets all variables when *any* objective fails to converge. A single combined MSE objective, preserving DA derivatives (no `CONS()` calls), allows the gradient-based algorithm to converge monotonically.
5. **Warm-starting across physics models.** Running FR 1 cold from default initial guesses fails (MSE ~ 0.2); warm-starting from the converged FR 0 solution reduces this to MSE $\sim 8 \times 10^{-8}$ with only minor current adjustments ($\Delta I_{\text{max}} = 0.76$ A).

Next steps.

- Obtain the correct scaling coefficient for the chicane dipole field map to enable MGE-based simulation (the most physically accurate dipole model available in COSY).
- Add η'_x or $\eta_x(117)$ objective to control undulator dispersion.
- Constrain Stage 5 variables to $I > 0$ (e.g., via I^2 parameterisation) to find the positive-polarity COSY solution.

References

- [1] M. Weinberg, A. Fisher, and R. Li, “Design of the UH Mark V Free Electron Laser,” arXiv:2510.14061v1, Table I.